

E-commerce Sales SQL Project

This document contains a complete SQL project for E-commerce Sales analysis. It includes a dataset of 1,000 rows and 20 SQL queries with answers. The project covers sales, customer behavior, revenue trends, and advanced SQL techniques.

1. Total revenue

```
SELECT SUM(revenue) AS total_revenue FROM orders;
```

2. Total number of orders

```
SELECT COUNT(DISTINCT order_id) AS total_orders FROM orders;
```

3. Total revenue per category

```
SELECT category, SUM(revenue) AS category_sales FROM orders GROUP BY category ORDER BY category_sales DESC;
```

4. Top 5 products by revenue

```
SELECT product_id, SUM(revenue) AS product_sales FROM orders GROUP BY product_id ORDER BY product_sales DESC LIMIT 5;
```

5. Top 5 customers by spending

```
SELECT customer_id, SUM(revenue) AS total_spent FROM orders GROUP BY customer_id ORDER BY total_spent DESC LIMIT 5;
```

6. Average order value (AOV)

```
SELECT ROUND(AVG(revenue), 2) AS avg_order_value FROM orders;
```

7. Monthly sales trend

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month, SUM(revenue) AS monthly_sales FROM orders GROUP BY month ORDER BY month;
```

8. Region with highest revenue

```
SELECT region, SUM(revenue) AS region_sales FROM orders GROUP BY region ORDER BY region_sales DESC LIMIT 1;
```

9. Most popular category (by quantity sold)

```
SELECT category, SUM(quantity) AS total_qty FROM orders GROUP BY category ORDER BY total_qty DESC LIMIT 1;
```

10. Customers with more than 5 orders

```
SELECT customer_id, COUNT(DISTINCT order_id) AS order_count FROM orders GROUP BY customer_id HAVING COUNT(DISTINCT order_id) > 5;
```

11. Average revenue per customer

```
SELECT customer_id, ROUND(AVG(revenue), 2) AS avg_revenue FROM orders GROUP BY customer_id;
```

12. Highest single order value

```
SELECT order_id, MAX(revenue) AS max_order_value FROM orders;
```

13. Unique customers per region

```
SELECT region, COUNT(DISTINCT customer_id) AS unique_customers FROM orders GROUP BY region;
```

14. Cumulative revenue over time

```
SELECT order_date, SUM(revenue) OVER (ORDER BY order_date) AS cumulative_revenue FROM orders;
```

15. Percentage contribution of each category

```
SELECT category, ROUND(SUM(revenue) * 100.0 / (SELECT SUM(revenue) FROM orders), 2) AS pct_sales FROM orders GROUP BY category;
```

16. Top 3 categories by average order value

```
SELECT category, ROUND(AVG(revenue), 2) AS avg_order_value FROM orders GROUP BY category ORDER BY avg_order_value DESC LIMIT 3;
```

17. Customer with highest total quantity purchased

```
SELECT customer_id, SUM(quantity) AS total_quantity FROM orders GROUP BY customer_id ORDER BY total_quantity DESC LIMIT 1;
```

18. Total revenue per year

```
SELECT YEAR(order_date) AS year, SUM(revenue) AS yearly_sales FROM orders GROUP BY YEAR(order_date) ORDER BY year;
```

19. Repeat customers

```
SELECT customer_id, COUNT(DISTINCT order_id) AS order_count FROM orders GROUP BY customer_id HAVING COUNT(DISTINCT order_id) > 1;
```

20. Day with maximum revenue

```
SELECT order_date, SUM(revenue) AS daily_sales FROM orders GROUP BY order_date ORDER BY daily_sales DESC LIMIT 1;
```